

show that the set of all real valued polynomials of degree less than or equal to n denoted by $P_n(\mathbb{R})$ is a vector space with operations of addition & scalar multiplication.
Given:

$$\text{Let } V = P_n(\mathbb{R}) \quad p(x) = \left\{ a_0 + a_1x + a_2x^2 + \dots + a_nx^n \mid a_i \in \mathbb{R} \right\}$$
$$= \sum_{i=0}^n a_i x^i$$

$$q(x) = \left\{ b_0 + b_1x + b_2x^2 + \dots + b_nx^n \mid b_i \in \mathbb{R} \right\}$$
$$= \sum_{i=0}^n b_i x^i$$

$$\text{Defn } r(x) = \sum_{i=0}^n c_i x^i$$

VS1: closure property:

To prove: $p(x) + q(x) \in V$

$$\begin{aligned} p(x) + q(x) &= \sum_{i=0}^n a_i x^i + \sum_{i=0}^n b_i x^i \\ &= \sum_{i=0}^n (a_i + b_i) x^i \in V \end{aligned}$$

$\therefore$ It is satisfied.

VS2: Associative property:

Let $p(x), q(x), r(x) \in V$

To prove: $p(x) + [q(x) + r(x)] = [p(x) + q(x)] + r(x)$

LHS: $p(x) + [q(x) + r(x)]$

$$= \sum_{i=0}^n a_i x^i + \sum_{i=0}^n (b_i + c_i) x^i$$

$$= \sum_{i=0}^n (a_i + b_i) x^i + \sum_{i=0}^n c_i x^i$$

$$= [p(x) + q(x)] + r(x)$$

= RHS $\therefore$ It is verified.

VS3: Identity property

$\forall p(x) \in V, \exists 0(x) \in V$

$$p(x) + 0(x) = \sum_{i=0}^n a_i x^i + 0$$

$$= \sum_{i=0}^n a_i x^i$$

$$= p(x) \in V$$

$\therefore$ It is verified.

VS4: Inverse property:

$\forall p(x) \in V, \exists -p(x) \in V$

$$p(x) + (-p(x)) = \sum_{i=0}^n a_i x^i + \left(-\sum_{i=0}^n a_i x^i\right)$$

$$= \sum_{i=0}^n (a_i x^i - a_i x^i)$$

$$= 0$$

$\therefore$ It is satisfied

V55: commutative property

let $p(x), q(x) \in V$, then

To prove: $p(x) + q(x) = q(x) + p(x)$

LHS: $p(x) + q(x)$

$$= \sum_{i=0}^n a_i x^i + \sum_{i=0}^n b_i x^i$$

$$= \sum_{i=0}^n (a_i + b_i) x^i$$

$$= \sum_{i=0}^n (b_i + a_i) x^i$$

$$= \sum_{i=0}^n b_i x^i + \sum_{i=0}^n a_i x^i$$

$$= q(x) + p(x) \in V$$

$\therefore$ It is satisfied.

V56: Closure property over scalar multiplication

let $\alpha \in \mathbb{R}$, $\exists p(x) \in V$

To prove: $\alpha p(x) \in V$

$$\text{Now, } \alpha p(x) = \alpha \sum_{i=0}^n a_i x^i$$

$$= \alpha a_i x^i \in V$$

$\therefore$ It is satisfied.

V57: $\forall p(x) \in V$

To prove: $1 \cdot p(x) = p(x)$

$$\text{let } 1 \cdot p(x) = 1 \cdot \sum_{i=0}^n a_i x^i$$

$$= \sum_{i=0}^n a_i x^i \in V$$

$\therefore$ It is satisfied.

V88: Let $\alpha, \beta \in \mathbb{R}$, $p(x) \in V$ then

To prove:

$$(\alpha\beta) p(x) = \alpha [\beta p(x)]$$

LHS:

$$(\alpha\beta) p(x) = (\alpha\beta) \sum_{i=0}^n a_i x^i$$

$$= \alpha \cdot \sum_{i=0}^n \beta a_i x^i$$

$$= \alpha [\beta p(x)] \in V$$

= RHS

$\therefore$ It is satisfied

V89: Distributive property of scalar multiplication over scalar addition.

Let $\alpha, \beta \in \mathbb{R}$, $\forall p(x) \in V$.

To prove: $(\alpha+\beta) p(x) = \alpha p(x) + \beta p(x)$

$$(\alpha+\beta) p(x) = (\alpha+\beta) \left(\sum_{i=0}^n a_i x^i \right)$$

$$= \alpha \sum_{i=0}^n a_i x^i + \beta \sum_{i=0}^n a_i x^i$$

$$= \alpha p(x) + \beta p(x) \in V$$

= RHS

$\therefore$ It is satisfied.

V90: Distributive property of scalar multiplication over vector addition.

Let $\alpha \in \mathbb{R}$, $\forall p(x), q(x) \in V$

To prove:

$$\alpha [p(x) + q(x)] = \alpha p(x) + \alpha q(x)$$

$$\alpha [p(x) + q(x)]$$

$$= \alpha (a_i x^i + b_i x^i)$$

$$= \alpha \sum_{i=0}^n a_i x^i + \sum_{i=0}^n b_i x^i$$

$$= \alpha p(x) + \alpha q(x)$$

$$= \text{RHS} \in V$$

$\therefore$ It is satisfied.

$\therefore$ It is satisfied all 10 property.

$\therefore$ Hence V is a vector space.

$\therefore$ Hence proved.